ENHANCING CYTOLOGICAL STAINING-FREE IMAGE CLASSIFICATION WITH ROBUST VISION TRANSFORMERS

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Introduction

ViT Over CNNs: Uses a DINO-DeiT-III Vision Transformer architecture to capture global cellular context and long-range dependencies.

Dissecting Complex Shapes: Models use sharpened attention weights to isolate subtle features, managing the extreme cell shape, size, and texture variations found in urine cytology.

Overcoming Imbalance & Variance: Integrates Asymmetric Loss (ASL) to handle severe class imbalance and employs data augmentation to mitigate overfitting from limited, highly variable cytology datasets.

Visual Transformers

DINO Self-Supervised Learning: Pairs a student network with a momentum-updated teacher network.

Paradigm: The student minimizes cross-entropy loss between different augmented views of the same image, using centering and sharpening to learn rich features and prevent training collapse.

DeiT-III Backbone & SERF/ASL Upgrades: SERF activation function for gradient stability, while ASL is integrated to handle high class imbalance.

Urine Cytology

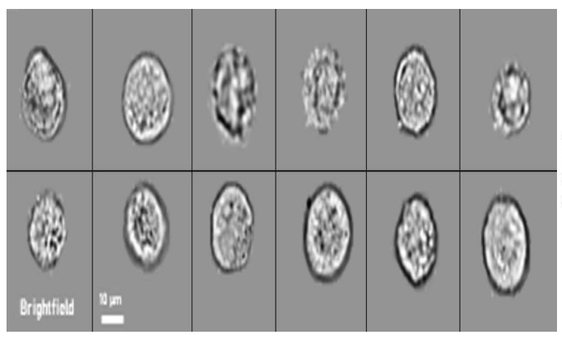
Non-Invasive Diagnostics: CELLo uses staining-free urine cell images to provide an early, non-invasive alternative to invasive cystoscopy.

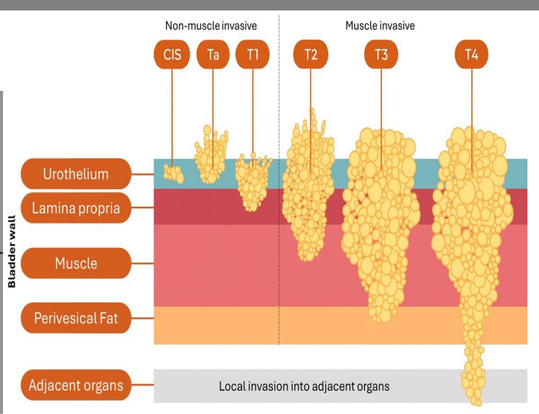
Image Characteristics: Consists of noisy, low-contrast 60× magnified greyscale images that require robust feature extraction to handle high cellular heterogeneity.

Classification & Grouping: Single-cell images are labeled by health status (Na), stage (Ta–T4), and grade (LG/HG), and are grouped by patient to prevent data leakage.

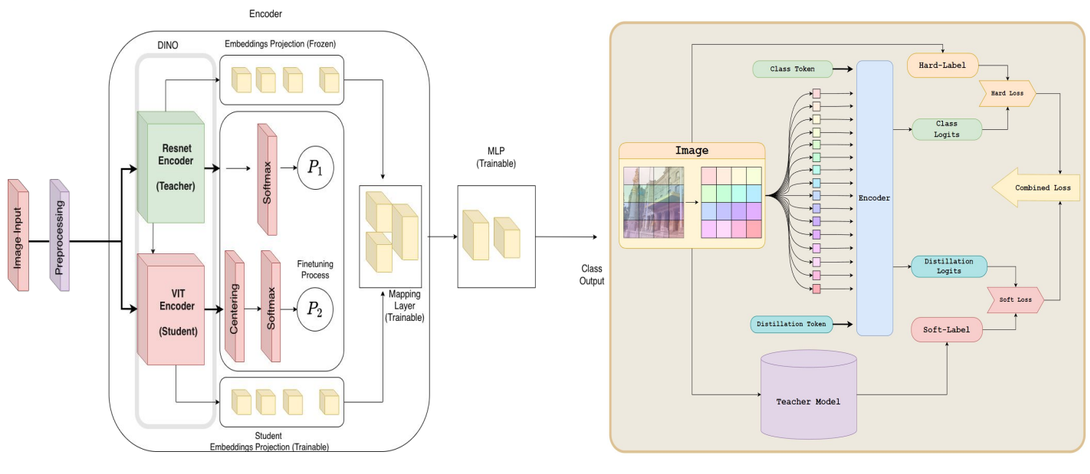
Implementation and Results

Urine **Cytology** and Cello





Proposed Deit III - DINO Model



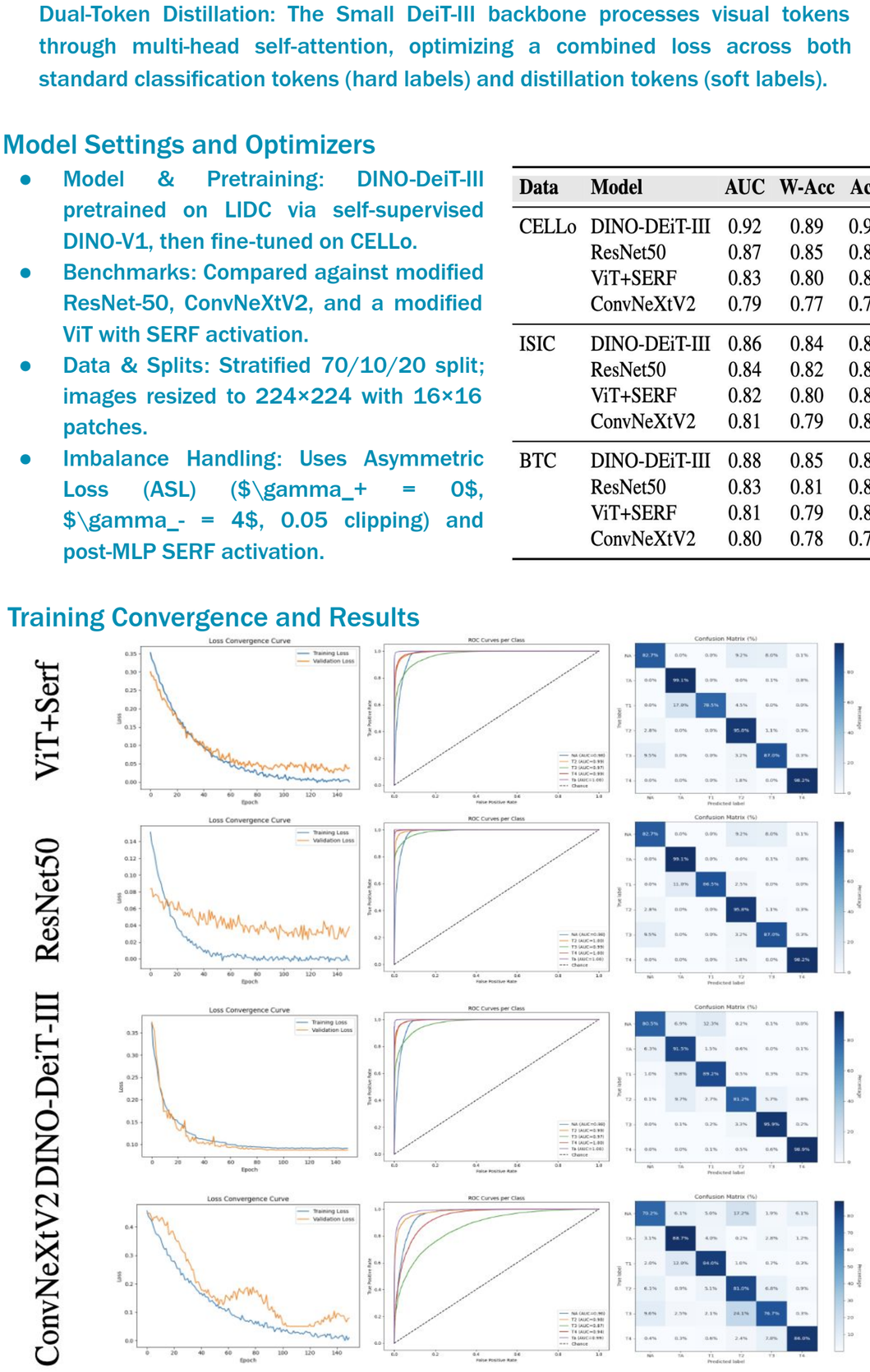
Dual-Token Distillation: The Small DeiT-III backbone processes visual tokens through multi-head self-attention, optimizing a combined loss across both standard classification tokens (hard labels) and distillation tokens (soft labels).

Model Settings and Optimizers

- Model & Pretraining: DINO-DeiT-III pretrained on LIDC via self-supervised DINO-V1, then fine-tuned on CELLo.
- Benchmarks: Compared against modified ResNet-50, ConvNeXtV2, and a modified ViT with SERF activation.
- Data & Splits: Stratified 70/10/20 split; images resized to 224×224 with 16×16 patches.
- Imbalance Handling: Uses Asymmetric Loss (ASL) ($\gamma_+ = 0$, $\gamma_- = 4$, 0.05 clipping) and post-MLP SERF activation.

Data	Model	AUC	W-Acc	Acc
CELLo	DINO-DEiT-III	0.92	0.89	0.90
	ResNet50	0.87	0.85	0.86
	ViT+SERF	0.83	0.80	0.82
	ConvNeXtV2	0.79	0.77	0.78
ISIC	DINO-DEiT-III	0.86	0.84	0.85
	ResNet50	0.84	0.82	0.83
	ViT+SERF	0.82	0.80	0.81
	ConvNeXtV2	0.81	0.79	0.80
BTC	DINO-DEiT-III	0.88	0.85	0.86
	ResNet50	0.83	0.81	0.82
	ViT+SERF	0.81	0.79	0.80
	ConvNeXtV2	0.80	0.78	0.79

Training Convergence and Results



Conclusions & Future Work

Top-Tier Performance: The DINO-DeiT-III model outperformed all benchmarks across three datasets, achieving its highest score on the CELLo dataset with an AUC of 0.92 and 0.90 accuracy.

Component Synergy: Combining self-supervised DINO pretraining, SERF activation, and Asymmetric Loss (ASL) generated a +5.5 p.p. AUC boost over the standard supervised baseline.

Effective Imbalance Handling: DINO pretraining drove the largest single advancement (+3.1 p.p. AUC), while ASL mitigated severe data imbalance to increase balanced accuracy by +1.5 p.p.

Enhanced Spatial Awareness: The framework successfully captured long-range context and isolated subtle, overlapping cellular features in noisy, staining-free cytological images.

Future Generalization Focus: Upcoming research will focus on Multimodal and mitigating domain shift.

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